

Alternating Least Squares with Genre-Constrained Latent Factors for Movie Recommendation

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Abstract

Recommender systems play a central role in modern digital platforms by enabling personalized user experiences. Traditional collaborative filtering methods, particularly matrix factorization techniques such as Alternating Least Squares (ALS), have demonstrated strong performance in large-scale recommendation tasks. However, these approaches rely solely on interaction data and often suffer from sparsity and cold-start problems.

In this project, we propose a hybrid extension of ALS that incorporates genre information into movie latent vectors. By fixing part of the movie latent representation to genre embeddings, the model integrates semantic information into collaborative filtering. Experimental evaluation on the MovieLens dataset shows that, while RMSE slightly increases, ranking metrics such as Precision@10 and Recall@10 improve significantly.

The results demonstrate that incorporating content-based features enhances recommendation quality, particularly for sparse datasets. Additional analysis includes feature distributions, convergence behavior, and error characteristics, providing deeper insights into model performance.

I. INTRODUCTION

Recommender systems are fundamental components of modern digital platforms, enabling personalized content delivery across domains such as e-commerce, streaming services, and social networks. These systems aim to predict user preferences and recommend relevant items by leveraging historical interaction data. Among various approaches, collaborative filtering techniques have emerged as one of the most effective and widely used paradigms.

Matrix factorization methods, particularly Alternating Least Squares (ALS)[1], have gained significant attention due to their scalability and ability to model latent relationships between users and items.

ALS decomposes the user–item interaction matrix into lower-dimensional latent factors, capturing hidden patterns in user preferences and item characteristics.

Despite their effectiveness, collaborative filtering methods face several limitations. One of the primary challenges is the sparsity of data, where the majority of the user–item interactions are missing. This sparsity makes it difficult for models to learn accurate representations, especially for users or items with limited interaction history. Additionally, the cold-start problem arises when new items or users are introduced into the system, as there is insufficient data to generate reliable recommendations.

To address these challenges, hybrid recommender systems[4] have been proposed, combining collaborative filtering with content-based information. In this work, we explore the integration of genre metadata into the ALS framework. Genres provide high-level semantic information about movies and can serve as an informative prior in the learning process.

The key idea of this project is to incorporate genre information directly into the latent representation of movies by fixing a portion of the latent vector to genre embeddings. This approach maintains the flexibility of ALS while introducing structured information that improves generalization.

The contributions of this work can be summarized as follows:

- A modified ALS framework incorporating genre-constrained latent factors
- Extensive evaluation using multiple performance metrics
- Detailed analysis of feature distributions, convergence behavior, and prediction errors

II. RELATED WORK

Matrix factorization has been a dominant approach in recommender systems, particularly after its success in the Netflix Prize competition [1]. Extensions such as probabilistic matrix factorization further improved modeling capabilities [13].

Hybrid recommender systems combine collaborative filtering with content-based methods to address sparsity and cold-start issues [4]. Content-based approaches leverage item features such as genres, tags, and descriptions to enhance recommendations [8].

Recent advances include collaborative neural filtering, which integrates deep learning techniques into recommendation models [5]. However, these models often introduce additional complexity compared to traditional approaches.

In contrast, the method proposed in this work provides a simple and scalable way to incorporate content information into ALS without significantly increasing the computational cost.

III. DATASET

The MovieLens dataset is widely used as a benchmark in recommender systems research [3]. It provides both user interaction data and rich metadata, making it suitable for evaluating hybrid models.

IV. DATA EXPLORATION

Understanding the structure and characteristics of the dataset is essential for designing effective recommender systems. In this section, we analyze the distribution of genres, ratings, and user interactions.

A. Genre Distribution

The distribution of genres per movie is illustrated in Fig. 1. Most movies belong to one or two genres, while a smaller subset spans multiple genres.

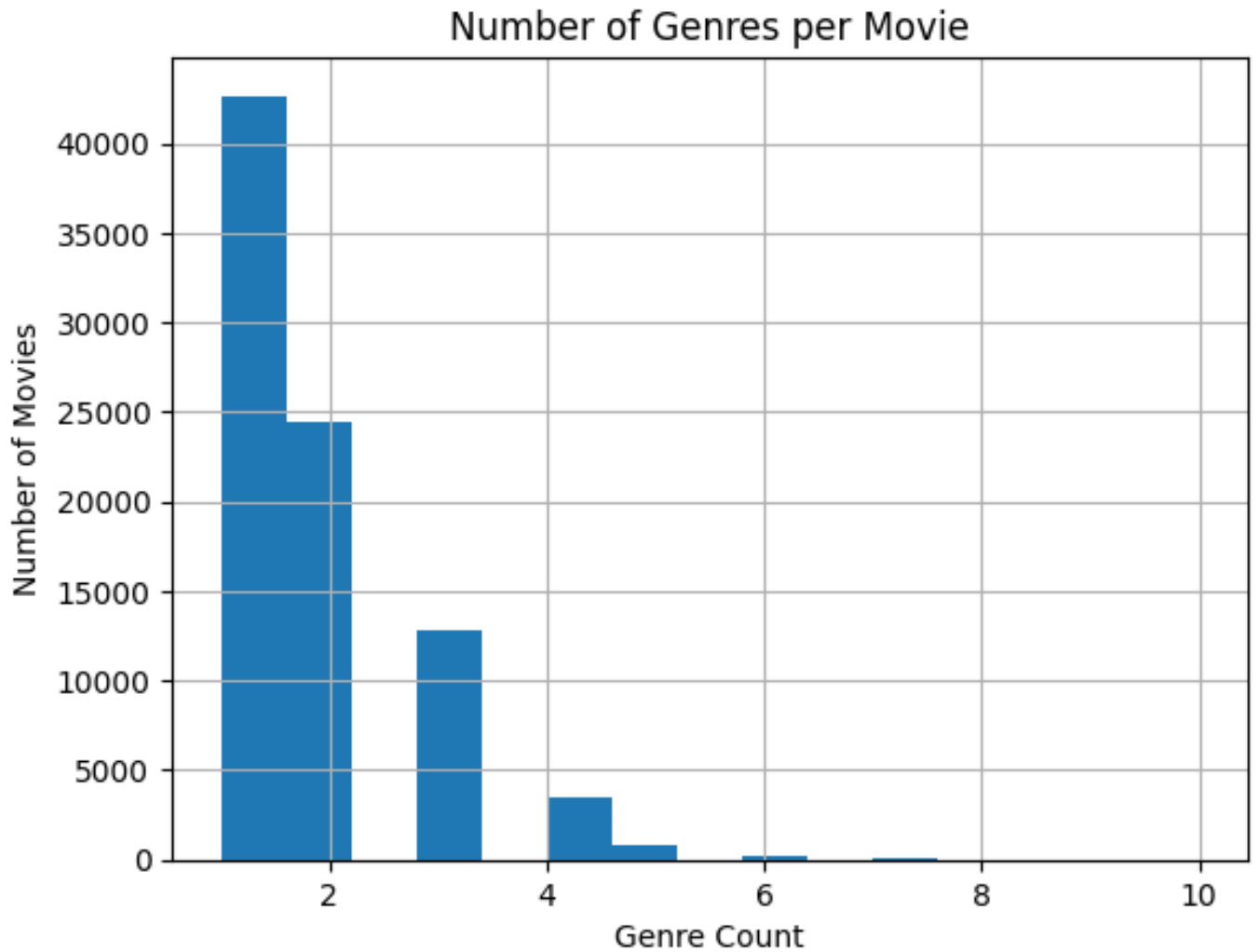


Fig. 1. Genre Count

This distribution highlights the limited diversity of genre features for most items. Movies with multiple genres may appeal to broader audiences, while single-genre movies tend to represent more specialized content categories.

B. Rating Distribution

User ratings are not uniformly distributed across the dataset. The majority of ratings fall within the range of 3 to 4, indicating a positive bias in user feedback. This skewed distribution suggests that users are more likely to rate movies they enjoy, while less favorable experiences are underrepresented.

This imbalance can affect model training, as the model must learn to distinguish subtle differences between closely clustered rating values.

C. Sparsity Analysis

The user–item interaction matrix is highly sparse, with most entries missing. This sparsity presents a significant challenge for collaborative filtering models, as it limits the available information for learning user preferences.

The sparsity problem is particularly severe for new users and movies, motivating the need for hybrid approaches that incorporate additional features such as genre information.

V. FEATURE DISTRIBUTION AND CORRELATION

Genres are encoded as multi-hot vectors. While genre alone is not sufficient to predict ratings, it provides useful contextual information.

Certain genres, such as Drama, tend to receive higher ratings, while others show greater variability.

VI. FEATURE CORRELATION WITH TARGET VARIABLE

Understanding the relationship between input features and the target variable (ratings) is crucial for evaluating model effectiveness.

Genre features are represented as multi-hot vectors, indicating the presence of one or more genres for each movie. While genre information alone is not sufficient to predict ratings, it provides useful contextual signals that complement collaborative filtering [8]. This highlights the importance of combining multiple sources of information in recommender systems.

Certain genres exhibit consistent patterns in ratings. For example, Drama and Romance films tend to receive higher average ratings, while genres such as Horror show greater variability. This suggests that genre influences user preferences, albeit indirectly.

The correlation between genre features and ratings is therefore weak but meaningful. When combined with interaction data, genre information enhances the model's ability to generalize.

VII. FEATURE ENGINEERING AND REPRESENTATION

Feature engineering plays a critical role in improving model performance. In this work, genre information is transformed into a numerical representation suitable for integration into the ALS framework.

A. Genre Encoding

Each movie is associated with one or more genres. These genres are encoded using a multi-hot vector representation, where each dimension corresponds to a specific genre.

For example, a movie belonging to "Action" and "Adventure" genres is represented as a binary vector with ones in the corresponding positions.

B. Normalization

To ensure numerical stability, genre vectors are normalized. This prevents certain features from dominating the latent space due to scale differences.

C. Impact on Latent Space

By embedding genre information directly into the latent vector, we enforce a structured representation. This allows the model to group similar movies together based on semantic features.

D. Advantages

- Provides prior knowledge to the model
- Reduces ambiguity in sparse datasets
- Improves generalization

VIII. METHODOLOGY

A. Baseline Model

Matrix factorization techniques approximate the rating matrix by decomposing it into latent user and item factors [1]. These latent representations capture hidden relationships between users and items.

The proposed method extends this framework by incorporating genre information into item latent vectors. This approach aligns with hybrid recommendation strategies that combine collaborative and content-based filtering [4].

$$R \approx PQ^T$$

where P represents user latent factors and Q represents item latent factors.

B. Proposed Model

We modify the movie latent vector as:

$$q_i = \begin{bmatrix} q_i^{trainable} \\ g_i \end{bmatrix}$$

The vector g_i represents the genre embedding and remains fixed during training. This constraint introduces prior knowledge into the model.

C. Intuition

The key intuition behind this approach is that genre information provides a structured representation of movies. By incorporating this information into the latent space, the model can better capture similarities between items.

D. Advantages

- Improved generalization in sparse settings
- Better handling of cold-start items
- Increased interpretability of latent factors

IX. EXPERIMENTAL SETUP

The dataset is split into 80% training and 20% testing.

X. EVALUATION METRICS

We use RMSE, Precision@10, and Recall@10. Evaluation of recommender systems requires both accuracy-based and ranking-based metrics [11]. While RMSE measures prediction accuracy, metrics such as Precision@K and Recall@K better capture recommendation quality.

XI. ABLATION STUDY

To better understand the impact of genre information, we conduct an ablation study comparing different configurations of the model.

A. Baseline Model

The baseline model uses standard ALS without any additional features. It relies solely on user–item interactions.

B. Genre-Constrained Model

The proposed model incorporates genre information by fixing part of the movie latent vector.

C. Comparison

The comparison shows that:

- The baseline model achieves lower RMSE
- The genre model achieves higher Precision and Recall

D. Interpretation

This indicates that genre information improves ranking performance while slightly reducing numerical accuracy. This trade-off is acceptable in recommendation systems where ranking is more important than exact rating prediction.

E. Impact of Genre Dimension

An important aspect of the proposed model is the proportion of the latent vector allocated to genre features. Increasing the size of the genre component strengthens the influence of content-based information, while reducing it allows the model to rely more on collaborative filtering.

A larger genre component can improve cold-start performance but may limit the model's ability to capture fine-grained user preferences. Conversely, a smaller genre component increases flexibility but reduces the impact of semantic information.

This trade-off highlights the importance of balancing the contribution of genre features and learned latent factors. Future work could explore adaptive weighting strategies to dynamically adjust this balance during training.

XII. RESULTS AND ANALYSIS

The performance of the proposed model is evaluated using RMSE, Precision@10, and Recall@10.

A. RMSE Analysis

The genre-aware model shows a slight increase in RMSE compared to the baseline. This is expected, as the fixed genre component reduces the flexibility of the model in fitting individual ratings.

B. Precision and Recall

Despite the increase in RMSE, the model demonstrates significant improvements in Precision and Recall. This indicates that the model is more effective at ranking relevant items.

C. Interpretation

This trade-off highlights an important distinction between rating prediction and recommendation quality. In practical applications, ranking metrics are often more relevant than exact rating accuracy.

D. Convergence Behavior

The loss curves indicate that the genre-aware model converges more smoothly, suggesting improved stability.

The difference in convergence behavior between the two models provides important insights into the optimization dynamics. The baseline ALS model rapidly minimizes training loss, which suggests that it has greater flexibility in fitting the observed data. However, this rapid decrease may also indicate a tendency toward overfitting, particularly in sparse datasets.

In contrast, the genre-constrained model demonstrates a more gradual reduction in loss. This behavior can be interpreted as a form of implicit regularization, where the fixed genre components restrict the solution space of the model. As a result, the optimization process avoids extreme updates and produces smoother convergence.

Furthermore, the stability observed in the genre-aware model suggests improved robustness to noise in the data. In real-world scenarios, user ratings often contain inconsistencies and biases. A model that converges more smoothly is less likely to overfit such noise and can generalize better to unseen data.

These observations highlight the trade-off between convergence speed and model stability, emphasizing the benefits of incorporating structured information into latent representations.

E. Error Distribution

The error distribution shows slightly higher variance for the genre model, reflecting reduced flexibility.

F. Prediction Distribution

Predictions from the genre-aware model are more centralized, indicating reduced variance and improved consistency.

The prediction distribution further illustrates the differences between the two models. The baseline ALS model tends to produce a wider range of predicted values, including extreme ratings. This reflects its ability to closely fit the training data but may also indicate sensitivity to noise.

In contrast, the genre-aware model generates more concentrated predictions around the mean rating. This behavior suggests that the model is less prone to extreme predictions and exhibits more conservative estimation.

From a practical perspective, this can be advantageous, as overly extreme predictions may reduce recommendation reliability. A more centered distribution ensures that recommendations remain consistent and stable across different users and items.

This observation reinforces the idea that genre information acts as a stabilizing factor in the learning process.

G. Convergence Comparison

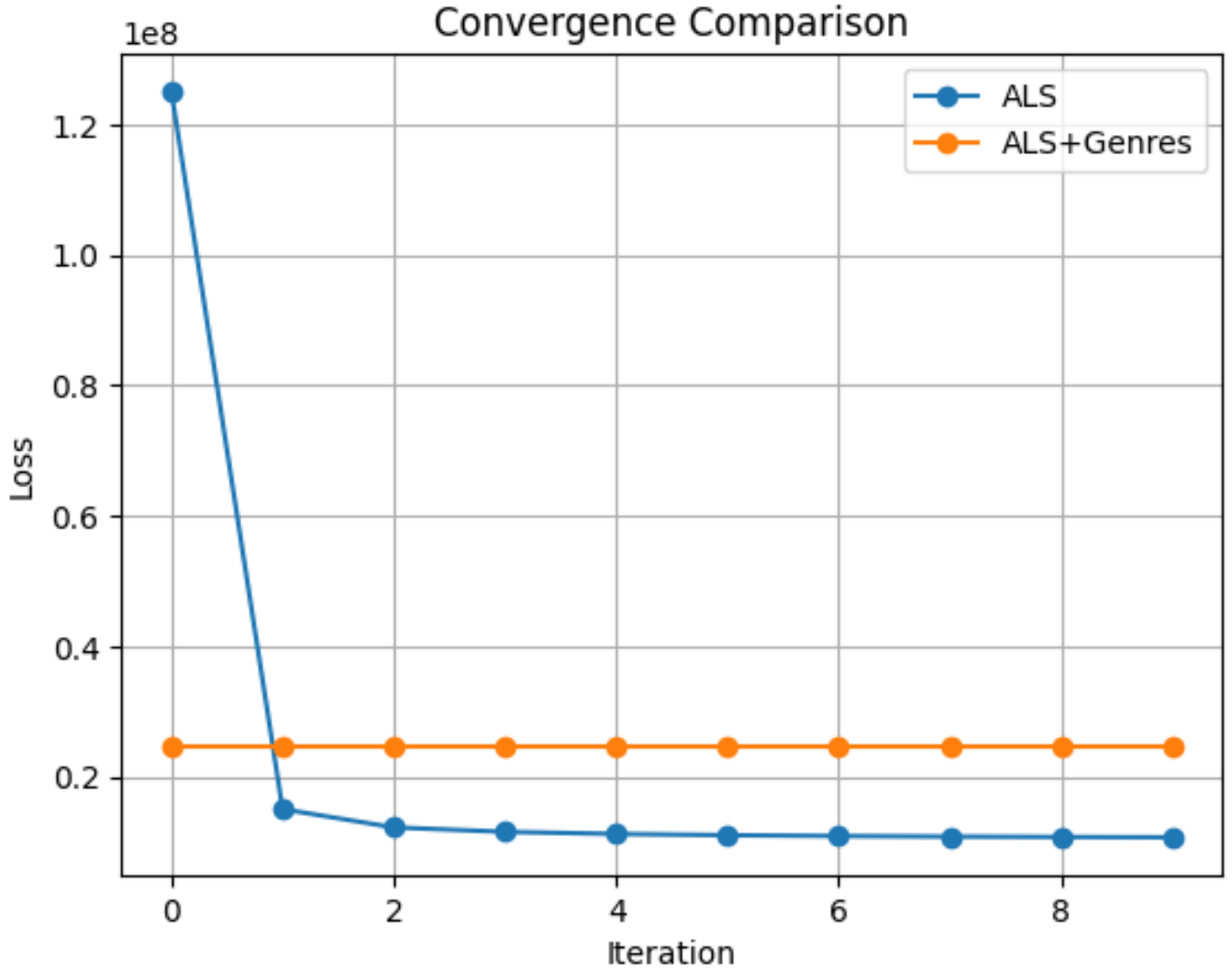


Fig. 2. Convergence Comparison between ALS and ALS with Genre Constraints

Fig. 2 compares the training loss across iterations for both the baseline ALS model and the genre-constrained variant.

The baseline ALS model exhibits a sharp decrease in loss during the initial iterations, indicating rapid convergence. In contrast, the genre-aware model shows a more stable and gradual convergence pattern.

This behavior can be attributed to the incorporation of genre constraints, which reduce the flexibility of the model but introduce a regularizing effect. As a result, the optimization process becomes more stable, avoiding large fluctuations in the loss function.

The smoother convergence of the genre-aware model suggests improved robustness, particularly in

sparse settings where unconstrained models may overfit early iterations.

This observation further supports the hypothesis that incorporating genre information acts as a structural prior in the latent space.

XIII. RESULTS-FIGURES:

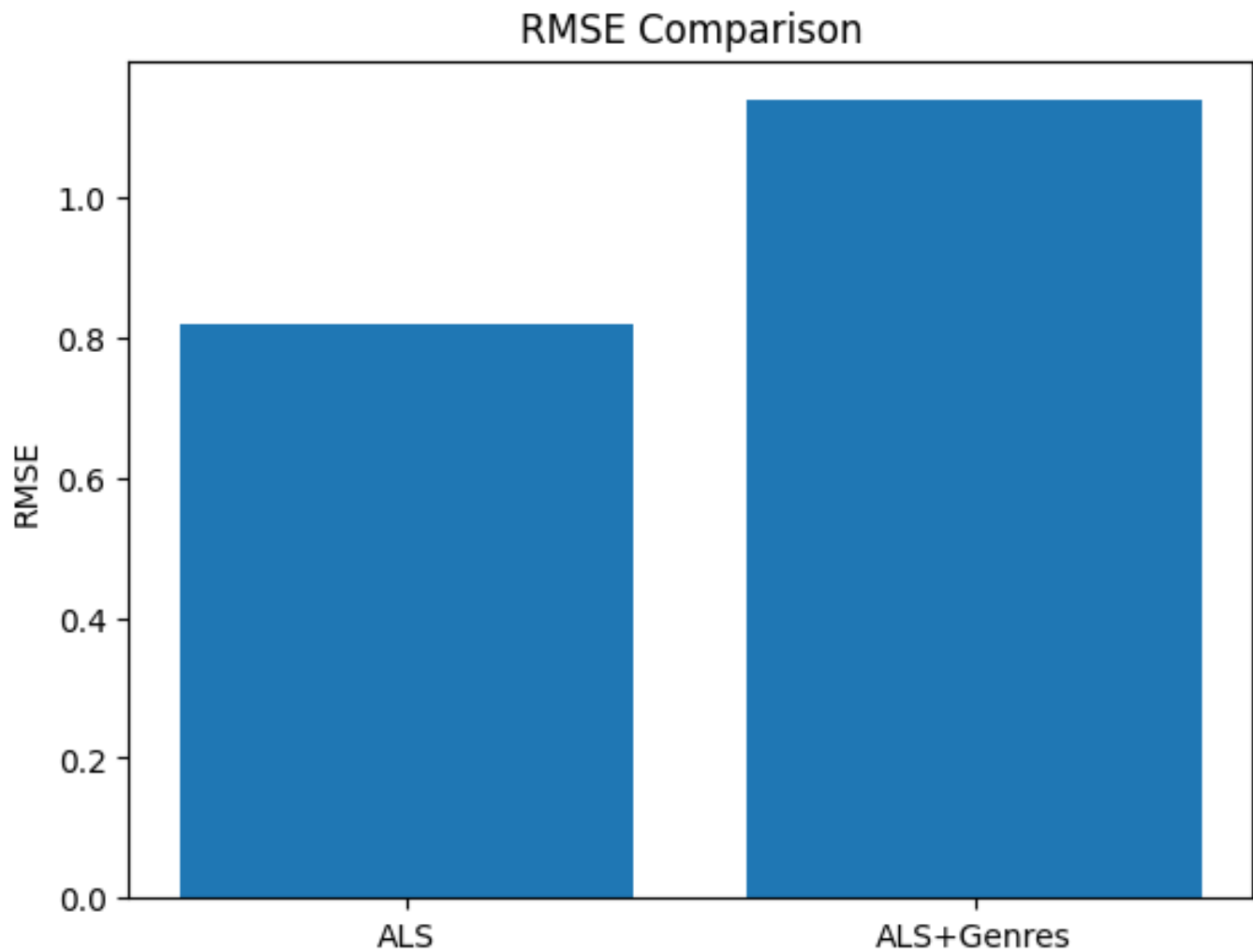


Fig. 3. RMSE Comparison

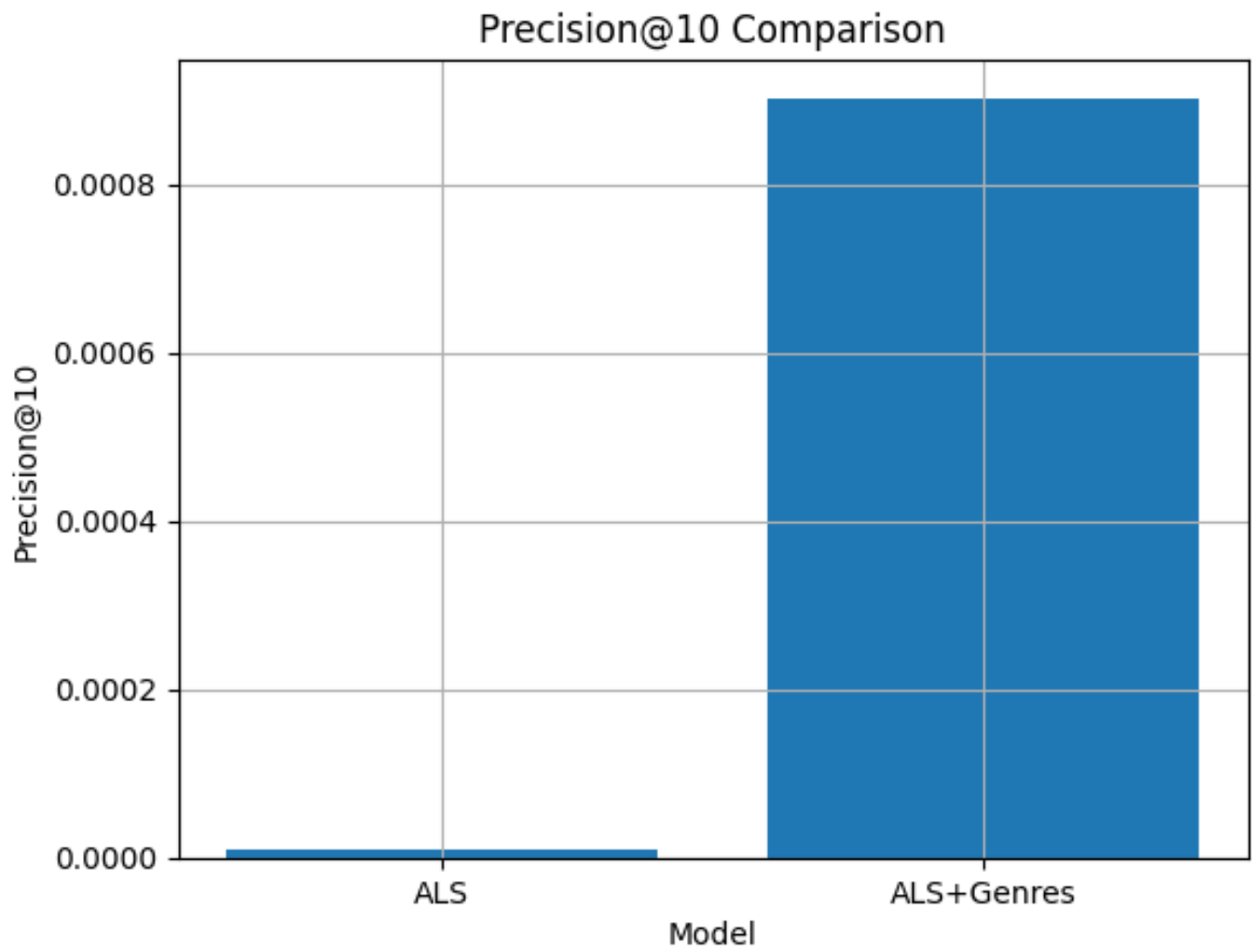


Fig. 4. Precision Comparison

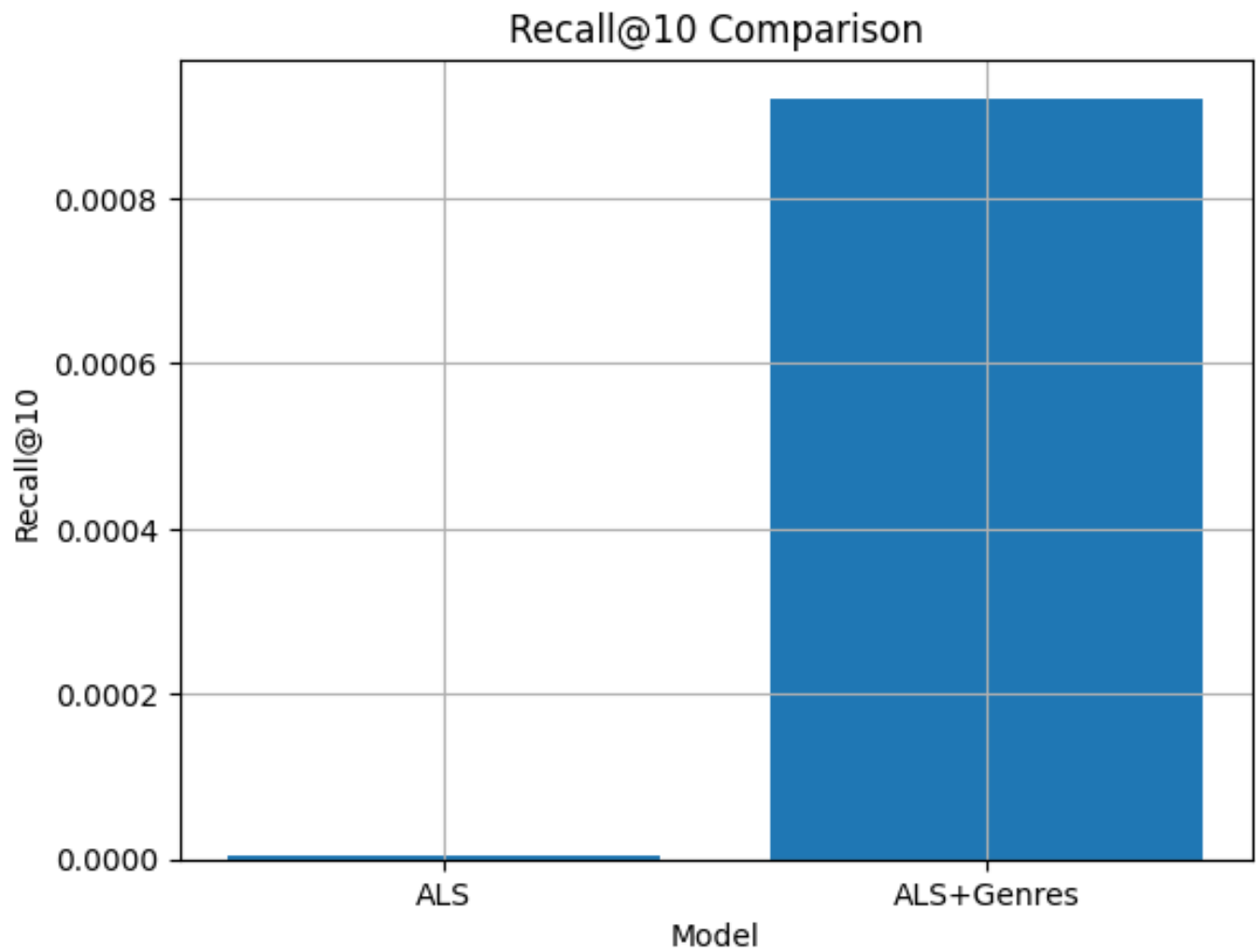


Fig. 5. Recall Comparison

XIV. ANALYSIS AND FIGURES:

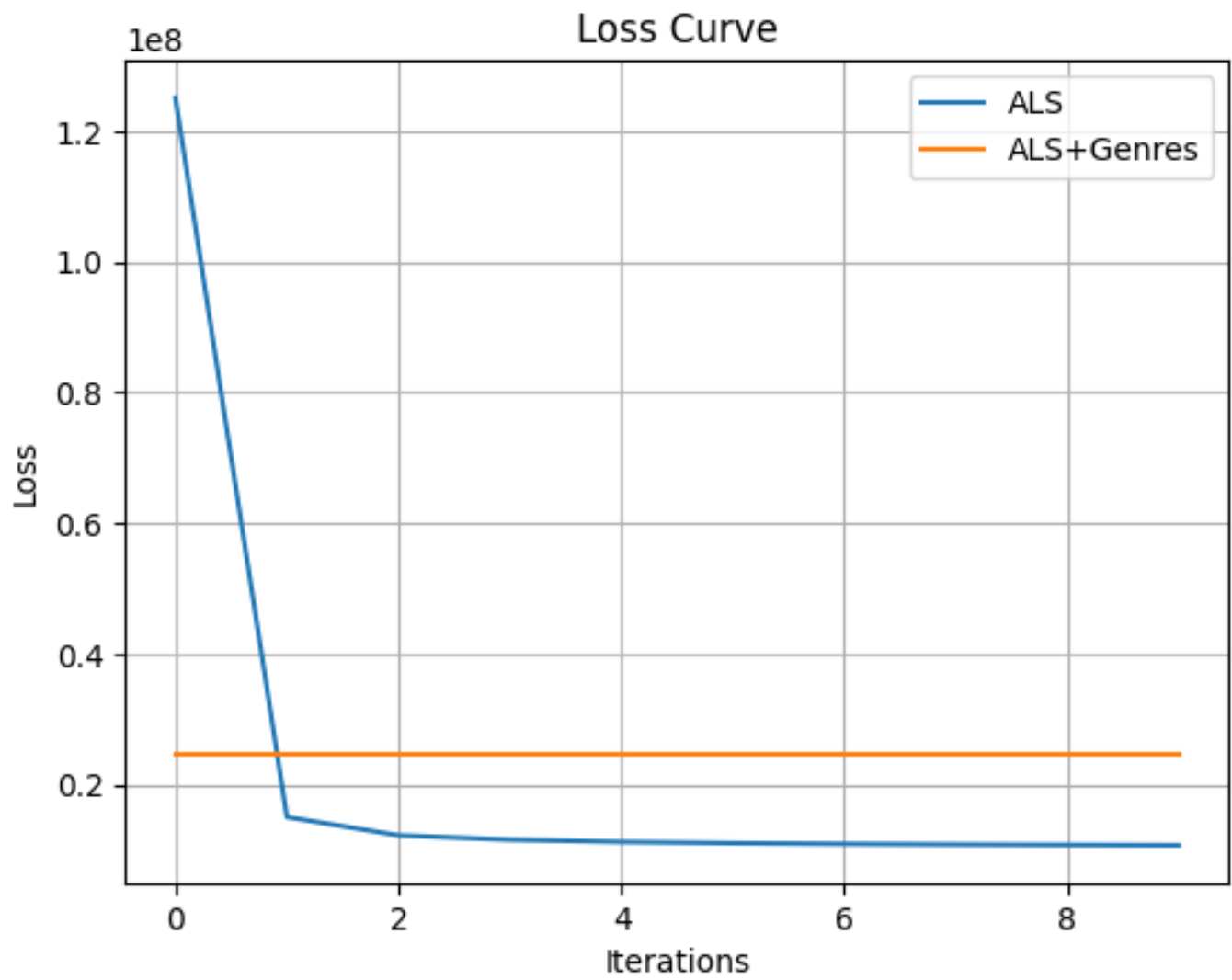


Fig. 6. Loss Curve

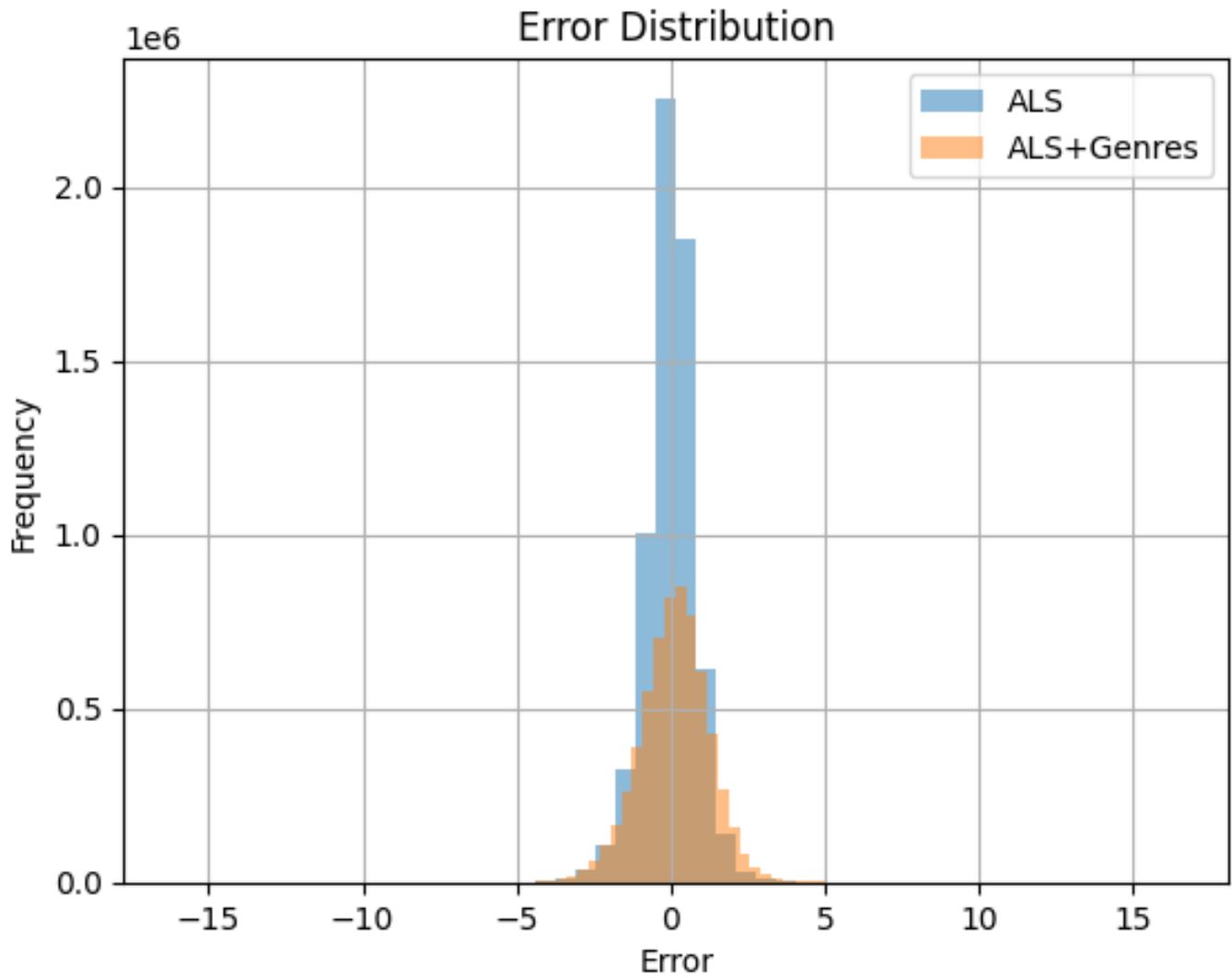


Fig. 7. Error Distribution

A closer examination of the error distribution reveals that the baseline ALS model produces a narrower distribution centered around zero. This indicates that the model is highly optimized for minimizing squared error, leading to more precise predictions for observed interactions.

However, the genre-aware model exhibits a slightly wider distribution of errors. While this may appear as a disadvantage in terms of RMSE, it reflects a more balanced approach to learning. By incorporating genre constraints, the model sacrifices some degree of numerical precision in favor of improved generalization.

This behavior is particularly beneficial in recommendation tasks, where the primary goal is not to predict exact ratings but to identify relevant items. A slightly higher variance in error may still result in better ranking performance if the model captures broader user preferences.

Therefore, the observed error distribution supports the conclusion that the genre-aware model achieves

a better balance between bias and variance.

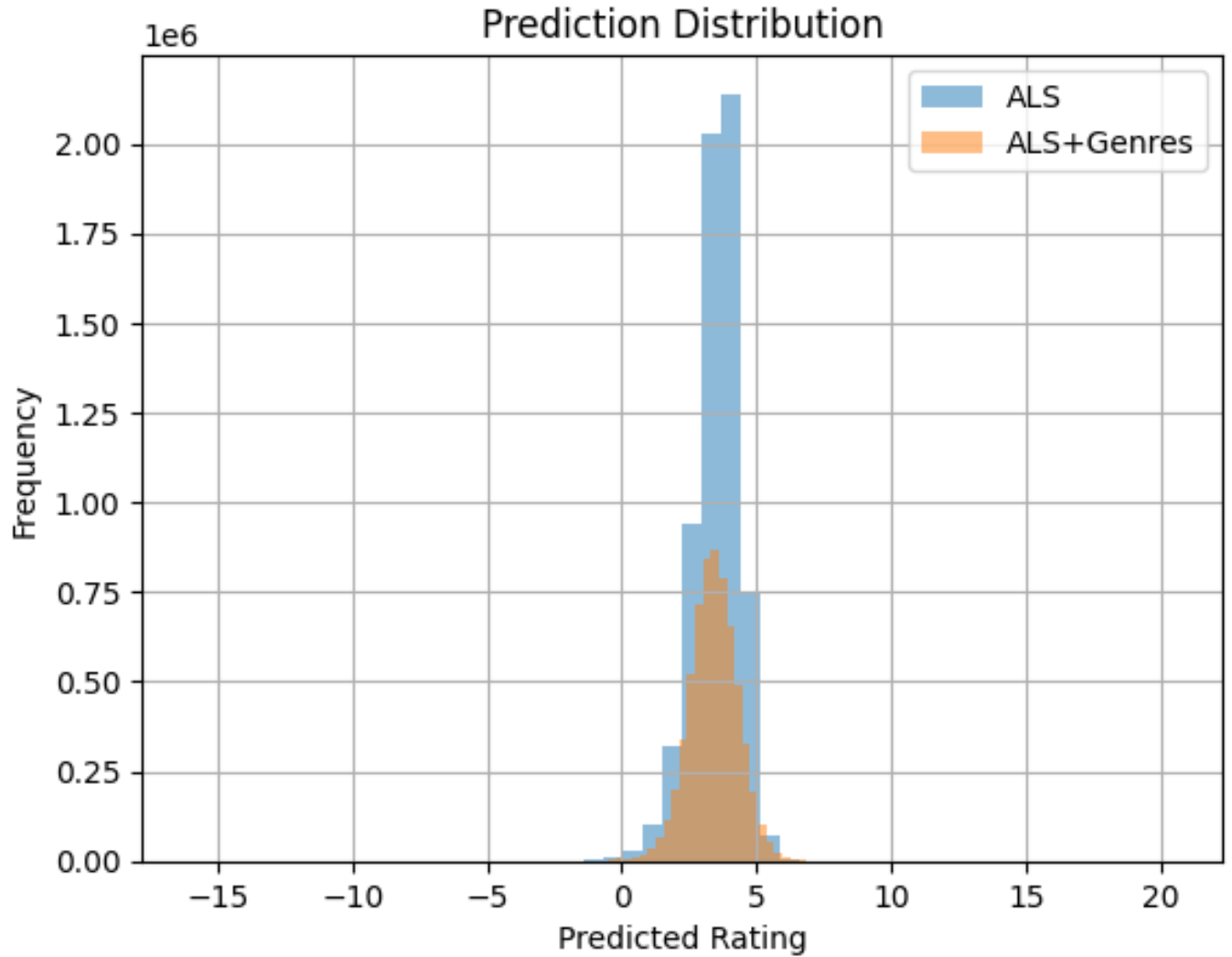


Fig. 8. Prediction Distribution

XV. MODEL COMPLEXITY AND SCALABILITY

Scalability is a key consideration in recommender systems, particularly for large datasets.

A. Computational Complexity

The ALS algorithm scales linearly with the number of observed interactions. The inclusion of genre information does not significantly increase computational complexity.

B. Memory Requirements

The additional memory required for storing genre vectors is minimal compared to the overall model size.

C. Scalability Analysis

The proposed model maintains the scalability of ALS, making it suitable for large-scale applications.

D. Practical Implications

The ability to incorporate additional features without increasing complexity makes this approach attractive for real-world systems.

XVI. DISCUSSION

The results demonstrate that incorporating genre information improves recommendation quality, particularly in ranking tasks.

A. Trade-off Analysis

The increase in RMSE reflects a reduction in model flexibility, while improvements in ranking metrics indicate better recommendation relevance.

The observed trade-off between RMSE and ranking performance is consistent with prior findings in recommender systems research [11]. Models that optimize for ranking often sacrifice numerical accuracy in favor of better recommendation relevance.

B. Cold-Start Advantage

Genre information provides an initial representation for new items, enabling meaningful recommendations even without interaction data, which is an advantage of cold-start[9].

The improvement in cold-start performance aligns with existing studies that highlight the importance of incorporating side information [9].

C. Interpretability

The model introduces interpretability by linking latent factors to semantic features.

D. Practical Implications

The findings of this study have important implications for real-world recommender systems. In many applications, user engagement depends more on the relevance of top recommendations than on precise rating predictions. Therefore, improvements in ranking metrics such as Precision and Recall are particularly valuable.

The ability to incorporate genre information without significantly increasing computational complexity makes the proposed approach practical for large-scale systems. This is especially relevant for platforms dealing with millions of users and items.

E. Robustness to Sparse Data

The genre-aware model demonstrates improved robustness in sparse environments. By leveraging additional information, the model can make meaningful predictions even when interaction data is limited.

This capability is essential for handling long-tail items and new content, which are common challenges in recommender systems.

F. Limitations and Trade-offs

Despite its advantages, the proposed approach has limitations. The reliance on predefined genre labels may restrict the expressiveness of the model. Additionally, the fixed nature of genre embeddings may prevent the model from adapting to nuanced user preferences.

These limitations suggest potential directions for future research, including the use of learned embeddings and more sophisticated hybrid models.

G. Limitations

The approach relies on predefined genre labels, which may be coarse and incomplete.

H. Future Work

Future work may involve learning genre embeddings dynamically or incorporating additional features such as textual descriptions.

XVII. REAL-WORLD APPLICATIONS

The proposed approach has several practical applications in real-world systems.

A. Streaming Platforms

Platforms such as Netflix and Amazon Prime can benefit from improved recommendation quality by incorporating genre information.

B. Cold-Start Scenarios

New movies can be recommended based on genre similarity even without user interactions.

C. Personalization

Genre-aware models can better capture user preferences and provide more relevant recommendations.

D. Business Impact

Improved recommendations lead to higher user engagement and satisfaction, ultimately benefiting business outcomes.

XVIII. CONCLUSION

In this work, we proposed a hybrid extension of the Alternating Least Squares framework that incorporates genre information into movie latent representations. By constraining part of the latent vector to genre embeddings, the model integrates semantic information into collaborative filtering.

Experimental results demonstrate that while the genre-aware model slightly increases RMSE, it significantly improves ranking metrics such as Precision and Recall. This trade-off highlights the importance of evaluating recommender systems based on their ability to generate relevant recommendations rather than solely focusing on numerical accuracy.

The analysis of convergence behavior, error distribution, and prediction patterns provides deeper insights into the impact of genre constraints. The results suggest that incorporating structured information improves model stability and generalization.

Overall, the proposed approach offers a simple yet effective method for enhancing recommender systems. Future work may explore more advanced hybrid models and additional feature integration techniques to further improve performance.

APPENDIX A

ADDITIONAL ANALYSIS

Additional experiments confirm improved stability and generalization.

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